

ColdCast Project Brief

Recommendation for the Predictive AI Evaluation Competition

Our recommendation We propose ColdCast, a deep learning system that predicts the probability that an AI system will answer a new evaluation item correctly without running that system on the item. This is a practical, competition-aligned project with clear baselines, measurable progress, and enough research depth for a strong semester contribution.

Problem and research question

Evaluating every model on every benchmark question is expensive, and average accuracy does not separate model ability from question difficulty. The competition asks us to estimate $P(\text{correct})$ for an unseen system-item response using model metadata, item content, prior response evidence, and a small label budget. **Research question:** Can semantic question representations, latent model capabilities, retrieval of similar items, and five-label adaptation produce better calibrated cold-start predictions than simple averages or standard neural baselines?

Proposed system and contribution

- **Represent each item** with a pretrained text encoder plus structured task and tool features.
- **Represent each AI system** with learned embeddings of model, provider, harness, version, and reasoning settings.
- **Retrieve relevant evidence** from semantically similar training items and comparable systems.
- **Adapt from five labels** by updating a small capability vector and comparing random, uncertainty-based, and diversity-based label selection.

Evidence plan Use benchmark-held-out validation to prevent leakage. Compare constant, global-rate, per-system-rate, metadata-only, text-only, retrieval-only, and neural factorization baselines. Report Brier score as the primary metric, with calibration error, reliability plots, cold-start performance, and ablations.

Why our team should choose it

ColdCast has a low-risk core and meaningful extensions. We can first deliver a text-plus-metadata predictor and calibration study, then add retrieval and active label selection. The work divides cleanly across data and validation, modeling, retrieval and adaptation, and experiments and reporting. It also requires less specialized infrastructure than EEG or neural-audio projects while still supporting a competition submission and a publishable research narrative.

Current studies and reading

1. [AIMS Predictive AI Evaluation Competition](#) — defines cold-start response prediction, small label budgets, and Brier-based evaluation.
2. [Truong et al. 2025, Reliable and Efficient Amortized Model-based Evaluation, ICML](#) — predicts item difficulty from content and uses adaptive testing to reduce evaluation cost.
3. [Shang, Tang, and He 2026, LLM Evaluation on Unseen Questions](#) — combines contextual embeddings with multidimensional item response theory; cross-scenario generalization remains an open problem.
4. [Jiang et al. 2026, Can We Trust Item Response Theory for AI Evaluation](#) — shows that small or nonstandard model samples can make IRT estimates unreliable, motivating neural and calibration checks.
5. [Zhuang et al. 2023, Efficiently Measuring the Cognitive Ability of LLMs](#) — applies computerized adaptive testing to estimate ability using fewer questions.

Team decision requested Approve ColdCast as our proposal direction and begin with data inspection and leakage-safe baselines